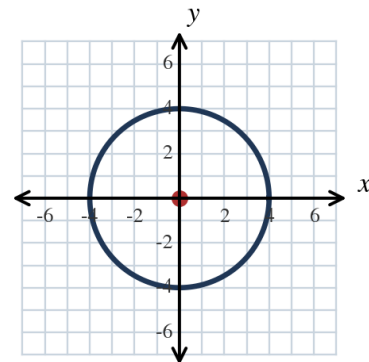


Speak Like A Mathematician:

standard equation of a circle:

where it comes from:



Circle Equations

Center (h, k), radius r: $(x - \underline{\hspace{1cm}})^2 + (y - \underline{\hspace{1cm}})^2 = \underline{\hspace{1cm}}$.

Center at the ORIGIN: $x^2 + y^2 = \underline{\hspace{1cm}}$.

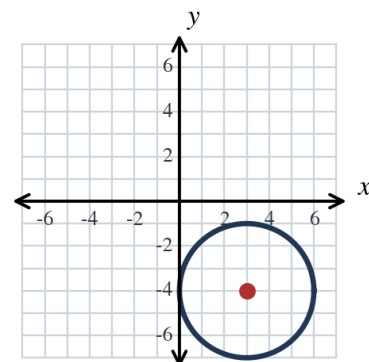
Watch the signs: $(x + 3)^2$ means $h = \underline{\hspace{1cm}}$.

Example 1 - From the Graph

- a) The top-right circle is centered at the origin with radius 4.
Write its equation.

Example 2 - From the Graph Again

- a) The circle at the right has center (3, -4) and radius 3.
Write its equation.



Example 3 - Is the Point on the Circle?

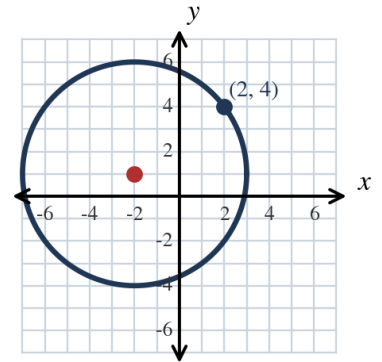
The circle at the right has center $(-2, 1)$ and radius 5. Point $(2, 4)$ is plotted.

- Write the equation of the circle.
- Show that $(2, 4)$ is on the circle.

Example 4 - Center and a Point

A circle has center $(1, -8)$ and passes through $(6, 4)$.

- Find the radius.
- Write the equation.



Example 5 - Graph It Yourself

- a) On the grid at the right, graph $(x - 1)^2 + (y + 2)^2 = 9$.
Label the center and one point on the circle.

Example 6 - On, Inside, or Outside?

Use the circle $(x - 2)^2 + (y + 2)^2 = 9$.

- a) Where is the point $(5, -2)$?
- b) How would you tell if a point is INSIDE?

Example 7 - From a Diameter

A diameter has endpoints $(-3, -4)$ and $(5, 2)$.

- a) Find the center.
- b) Find the radius and write the equation.

